Chapter 6 - TED Video

The TED chip reproduces the picture model of the C16 and Plus/4. It sits on compositor layer 12, between VGA (layer 10) and ANTIC (layer 13). TED's design folds video and audio into one chip; this chapter covers the video half. The audio half is described in Chapter 16.

6.1 What TED can show

Item	Value
Display area	320 × 200 pixels
Character grid	40 × 25 cells of 8 × 8
Border	32 left, 32 right, 35 top, 37 bottom
Total frame	384 × 272 pixels
Colours	121 unique (16 hues × 8 luminances minus duplicated blacks)
Modes	Text, bitmap, multicolour bitmap, extended-colour text

In multicolour mode each pixel is two screen pixels wide, so the effective resolution drops to 160 × 200 with four colours per cell instead of two.

6.2 BASIC keywords

The BASIC keyword `TED` introduces every TED subcommand. The recognised forms are:

Form	Effect
`TED ON`	Enable the chip (sets `TED_V_ENABLE` bit `0`).
`TED OFF`	Disable the chip.
`TED MODE`*n*	`0` = text, `1` = bitmap, `2` = multicolour bitmap. Sets the BMM and MCM bits and forces DEN on.
`TED COLOR`*bg*`[,`*border*`]`	Set background colour 0 and (optionally) the border colour.
`TED CHAR`*addr*	Set the character/bitmap base register.
`TED VIDEO`*addr*	Set the video matrix base register.
`TED CLS`	Fill the video matrix with the space character (`$20`) via the blitter.
`TED SCROLL`*dx*`,`*dy*	Write the X and Y fine-scroll fields of `CTRL2` and `CTRL1` (each `0`–`7`).

A minimal BASIC program that enables TED, paints the screen yellow on a blue background, and prints a banner through the video matrix:

```
10 TED ON
20 TED COLOR 6, 14         : REM bg=blue luminance 0, border=dark blue
30 TED CLS
40 REM (continue with POKE into the video matrix and colour RAM)
```

6.3 The register block

TED's video registers live in the small block `0xF0F20`–`0xF0F6B`. The chip's audio registers sit at `0xF0F00`–`0xF0F05` (Chapter 16).

Address	Name	Purpose
`0xF0F20`	`TED_V_CTRL1`	Control 1 (ECM, BMM, DEN, RSEL, YSCROLL).
`0xF0F24`	`TED_V_CTRL2`	Control 2 (RES, MCM, CSEL, XSCROLL).
`0xF0F28`	`TED_V_CHAR_BASE`	Character generator base (high nibble) and bitmap base (low nibble), in 1 KB steps.
`0xF0F2C`	`TED_V_VIDEO_BASE`	Video matrix base, in 1 KB steps (bits `3`–`7`).
`0xF0F30`	`TED_V_BG_COLOR0`	Background colour 0.
`0xF0F34`	`TED_V_BG_COLOR1`	Background colour 1 (multicolour).
`0xF0F38`	`TED_V_BG_COLOR2`	Background colour 2 (multicolour).
`0xF0F3C`	`TED_V_BG_COLOR3`	Background colour 3 (multicolour).
`0xF0F40`	`TED_V_BORDER`	Border colour.
`0xF0F44`	`TED_V_CURSOR_HI`	Cursor position, high byte.
`0xF0F48`	`TED_V_CURSOR_LO`	Cursor position, low byte.
`0xF0F4C`	`TED_V_CURSOR_CLR`	Cursor colour.
`0xF0F50`	`TED_V_RASTER_LO`	Current raster line, low byte (read-only).
`0xF0F54`	`TED_V_RASTER_HI`	Raster bit `8` in bit `0` (read-only).
`0xF0F58`	`TED_V_ENABLE`	Bit `0` = video enable.
`0xF0F5C`	`TED_V_STATUS`	Bit `0` = VBlank active (read-only).
`0xF0F60`	`TED_V_RASTER_CMP_LO`	Raster compare line, low byte.
`0xF0F64`	`TED_V_RASTER_CMP_HI`	Raster compare line bit `8` in bit `0`.
`0xF0F68`	`TED_V_RASTER_STATUS`	Bit `7` = compare pending; write `1` to clear.

Every register is a 32-bit word at a 4-byte-aligned address; only the low byte of each is meaningful.

6.3.1 CTRL1 bits

Bit	Name	Meaning
6	`ECM`	Extended colour mode.
5	`BMM`	Bitmap mode.
4	`DEN`	Display enable. Must be set for any picture.
3	`RSEL`	`0` = 24 rows, `1` = 25 rows.
0–2	`YSCROLL`	Vertical fine-scroll, `0`–`7` raster lines.

6.3.2 CTRL2 bits

Bit	Name	Meaning
5	`RES`	Reset.
4	`MCM`	Multicolour mode.
3	`CSEL`	`0` = 38 columns, `1` = 40 columns.
0–2	`XSCROLL`	Horizontal fine-scroll, `0`–`7` pixels.

The picture mode is the combination of `BMM`, `MCM`, and `ECM`:

`BMM`	`MCM`	`ECM`	Picture
0	0	0	Standard text.
0	1	0	Multicolour text.
0	0	1	Extended-colour text (8 distinct background colours from the top three bits of the character code).
1	0	0	Standard bitmap.
1	1	0	Multicolour bitmap (`160` × `200`).

6.4 The colour byte

Every colour register and every byte of colour RAM holds an 8-bit **colour byte** with this layout:

```
bit 7 6 5 4 3 2 1 0
    0 L L L H H H H
      |-luminance| |---hue---|
```

`L` is the luminance (`0`–`7`); `H` is the hue (`0`–`15`). The hue table is:

Hue	Name	Hue	Name
`0`	Black	`8`	Orange
`1`	White	`9`	Brown
`2`	Red	`10`	Yellow-green
`3`	Cyan	`11`	Pink
`4`	Purple	`12`	Blue-green
`5`	Green	`13`	Light blue
`6`	Blue	`14`	Dark blue
`7`	Yellow	`15`	Light green

Hue `0` (black) is the same colour at every luminance, which is why TED has `121` unique colours rather than `128` (16 × 8).

6.5 The VRAM region

TED owns 16 KB of private VRAM beginning at `0xF3000`. This region holds three things, all of which can be relocated through the `TED_V_VIDEO_BASE` and `TED_V_CHAR_BASE` registers:

Block	Default offset (from 0xF3000)	Size
Video matrix	0x0000	1024 B
Colour RAM	0x0400	1024 B
Character set	0x0800	2048 B

The video matrix is a 40 × 25 array of character codes. The colour RAM is a 40 × 25 array of colour bytes, one per cell. The character set is 256 entries of 8 bytes each; bit 7 of every byte is the leftmost pixel.

TED_V_VIDEO_BASE selects the video matrix base in 1 KB steps, through bits 3–7 of its low byte. TED_V_CHAR_BASE packs two selectors into one byte: bits 4–7 choose the character set, and bits 0–3 choose the bitmap base when in bitmap mode. Both selectors are also in 1 KB steps.

6.6 The cursor

The cursor is a single character cell. Its position is the 11-bit linear offset stored in TED_V_CURSOR_HI/LO (row * 40 + col), and its colour is in TED_V_CURSOR_CLR. The cursor blinks automatically at one cycle every 30 frames.

6.7 Raster interrupts

TED_V_RASTER_LO and bit 0 of TED_V_RASTER_HI report the current scan-line as a 9-bit value (0–311 over the full PAL frame). Writing the same 9-bit value into the corresponding compare registers arms the raster compare; when the raster reaches the line, TED_V_RASTER_STATUS bit 7 is set and the chip raises its interrupt line. The CPU clears the latch by writing 1 to bit 7 of TED_V_RASTER_STATUS.

Raster interrupts are how the C16 and Plus/4 split the screen into multiple regions with different scroll, colour, or character-set choices. The same trick works here.

6.8 Putting it together

The shortest TED program from machine language:

1. Write a non-zero value to TED_V_ENABLE (or use TED ON).
2. Set the picture mode in TED_V_CTRL1 and TED_V_CTRL2 (or use TED MODE).
3. Choose where the video matrix and character set live with TED_V_VIDEO_BASE and TED_V_CHAR_BASE.
4. Fill the video matrix and colour RAM in the regions you chose.
5. To animate, poll TED_V_STATUS bit 0 for the vertical retrace, or arm a raster compare.

The next chapter covers ANTIC and GTIA, the Atari 8-bit picture chips, which live on layer 13.